

SKILL-8

INPUT

```
GNU nano 8.7.1 builtin_commands.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

void show_help()
{
    printf("Built-in commands:\n");
    printf("help - Display help\n");
    printf("pwd - Display current directory\n");
    printf("cd - Change directory\n");
    printf("exit - Exit the shell\n");
}

int main()
{
    char command[100];

    printf("Mini Shell - Practical 8\n");
    printf("Type 'help' for commands.\n");

    while (1)
    {
        printf("shell> ");
        fflush(stdout);

        if (fgets(command, sizeof(command), stdin) == NULL)
            break;

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "help") == 0)
        {
            show_help();
        }
        else if (strcmp(command, "pwd") == 0)
        {
            char cwd[1024];

            if (getcwd(cwd, sizeof(cwd)) != NULL)
                printf("%s\n", cwd);
        }
        else if (strncmp(command, "cd ", 3) == 0)
        {
            if (chdir(command + 3) != 0)
                perror("cd");
        }
        else if (strcmp(command, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }
        else if (strlen(command) > 0)
        {
            printf("Invalid command: %s\n", command);
        }
    }

    return 0;
}
```



OUTPUT

```
administrator@VBCO-LAP-29:~$ mkdir practical8
cd practical8
pwd
/home/administrator/practical8
administrator@VBCO-LAP-29:~/practical8$ nano variable_expansion.c
administrator@VBCO-LAP-29:~/practical8$ gcc variable_expansion.c -o variable_expansion
administrator@VBCO-LAP-29:~/practical8$ ls
variable_expansion  variable_expansion.c
administrator@VBCO-LAP-29:~/practical8$ ./variable_expansion
Variable Expansion Test
=====

Hello $NAME -> Hello Hello Ubuntu
User: $USER -> User: administrator
Home: $HOME -> Home: /home/administrator
Unknown: $ABC -> Unknown: Unknown
administrator@VBCO-LAP-29:~/practical8$ nano builtin_commands.c

administrator@VBCO-LAP-29:~/practical8$ gcc builtin_commands.c -o builtin_commands
administrator@VBCO-LAP-29:~/practical8$ ls
builtin_commands  builtin_commands.c  variable_expansion  variable_expansion.c
administrator@VBCO-LAP-29:~/practical8$ ./builtin_commands
Mini Shell - Practical 8
Type 'help' for commands.
shell> help
Built-in commands:
help - Display help
pwd - Display current directory
cd - Change directory
exit - Exit the shell
shell> pwd
/home/administrator/practical8
shell> cd/tmp
Invalid command: cd/tmp
shell> pwd
/home/administrator/practical8
shell> cd /tmp
shell> pwd
/tmp
shell> cd /
shell> pwd
/
shell> hello
Invalid command: hello
shell> exit
Exiting shell...
administrator@VBCO-LAP-29:~/practical8$ ls -l
total 44
-rwxr-xr-x 1 administrator administrator 16520 Sep 27 16:26 builtin_commands
-rw-r--r-- 1 administrator administrator 1302 Sep 27 16:26 builtin_commands.c
-rwxr-xr-x 1 administrator administrator 16232 Sep 27 16:23 variable_expansion
-rw-r--r-- 1 administrator administrator 1200 Sep 27 16:23 variable_expansion.c
administrator@VBCO-LAP-29:~/practical8$ |
```